1840. Maximum Building Height

Solved

Hard Topics Hint

You want to build `n` new buildings in a city. The new buildings will be built in a line and are labeled from `1` to `n`.

However, there are city restrictions on the heights of the new buildings:

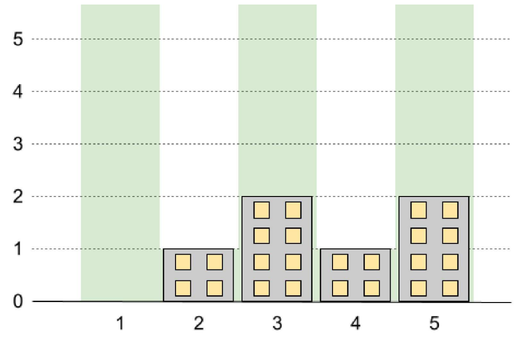
- The height of each building must be a non-negative integer.
- The height of the first building **must** be `0`.
- The height difference between any two adjacent buildings **cannot exceed** `1`.

Additionally, there are city restrictions on the maximum height of specific buildings. These restrictions are given as a 2D integer array `restrictions` where `restrictions[i] = [id_i, maxHeight_i]` indicates that building `id_i` must have a height **less than or equal to** `maxHeight_i`.

It is guaranteed that each building will appear **at most once** in `restrictions`, and building `1` will **not** be in `restrictions`.

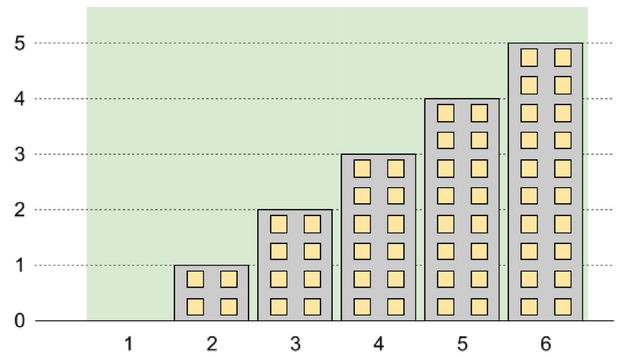
Return *the **maximum possible height** of the **tallest** building*.

Example 1:



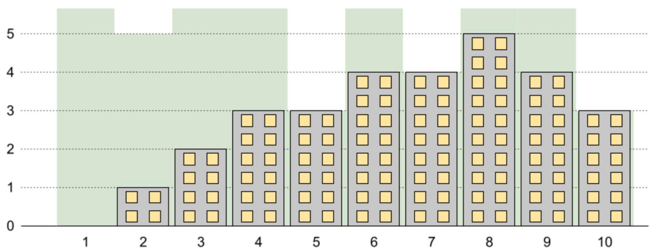
> **Input:** n = 5, restrictions = [[2,1],[4,1]]
> **Output:** 2
> **Explanation:** The green area in the image indicates the maximum allowed height for each building.
> We can build the buildings with heights [0,1,2,1,2], and the tallest building has a height of 2.

Example 2:



> **Input:** n = 6, restrictions = []
> **Output:** 5
> **Explanation:** The green area in the image indicates the maximum allowed height for each building.
> We can build the buildings with heights [0,1,2,3,4,5], and the tallest building has a height of 5.

Example 3:



> **Input:** n = 10, restrictions = [[5,3],[2,5],[7,4],[10,3]]
> **Output:** 5
> **Explanation:** The green area in the image indicates the maximum allowed height for each building.
> We can build the buildings with heights [0,1,2,3,3,4,4,5,4,3], and the tallest building has a height of 5.

Constraints:

- `2 <= n <= 10^9`
- `0 <= restrictions.length <= min(n - 1, 10^5)`
- `2 <= id_i <= n`
- `id_i` is **unique**.
- `0 <= maxHeight_i <= 10^9`

Seen this question in a real interview before? 1/6

Yes No

Accepted 78,457 /116.3K | Acceptance Rate 67.4 %

Topics

Senior Staff Array Math Sorting Weekly Contest 238

Hint 1

Hint 2

Similar Questions

Discussion (133)